

Name : Shan

Sap : 72387

BSAI

OOP (Lab)

◆ LAB Task 10

✧ Q1: Polymorphism Overloading Addition Methods.

➤ Input

Class EXC

```
package com.mycompany.overloading;

import java.util.Scanner;

public class Overloading {

    public static void main(String[] args) {

        int a,b,c;//shan,72387,BSAI

        Calculator C = new Calculator();

        Scanner s= new Scanner (System.in);//shan,72387,BSAI

        System.out.println("Enter values");

        a=s.nextInt();

        b=s.nextInt();//shan,72387,BSAI

        c=s.nextInt();

        int result = C.sum(a,b);//shan,72387,BSAI

        System.out.println("The sum of"+a+"and"+b+"is"+result);

        result = C.sum(a, b, c);//shan,72387,BSAI

        System.out.println("Sum of"+a+", "+b+", "+c+", "+result );

    }

}
```

Class Calculator

```
package com.mycompany.overloading;

public class Calculator {

    int sum (int a,int b,int c) {

        return a+b+c;//shan,72387,BSAI

    }

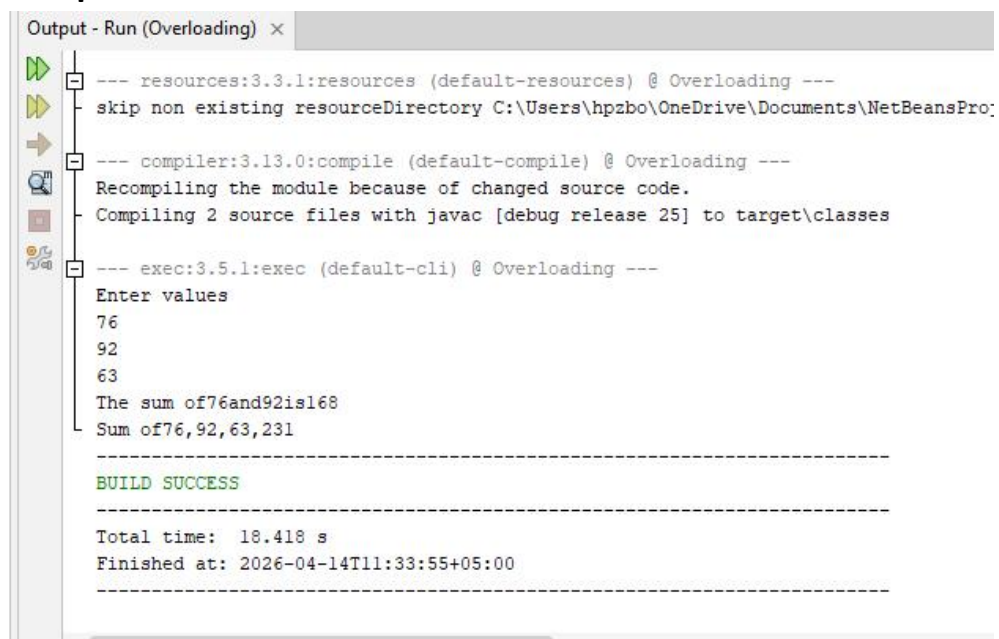
    int sum (int a,int b){

        return a+b;//shan,72387,BSAI

    }

}
```

➤ Output



```
Output - Run (Overloading) x
--- resources:3.3.1:resources (default-resources) @ Overloading ---
skip non existing resourceDirectory C:\Users\hpzbo\OneDrive\Documents\NetBeansPro
--- compiler:3.13.0:compile (default-compile) @ Overloading ---
Recompiling the module because of changed source code.
Compiling 2 source files with javac [debug release 25] to target\classes
--- exec:3.5.1:exec (default-cli) @ Overloading ---
Enter values
76
92
63
The sum of76and92isl68
Sum of76,92,63,231
-----
BUILD SUCCESS
-----
Total time:  18.418 s
Finished at: 2026-04-14T11:33:55+05:00
-----
```

✧ Q2: Polymorphism Overriding,Shape,Circle,Cylinder,etc.

➤ Input

Class Shapes

```
package com.mycompany.shapes.project;

public class Shape {

    void draw() {

        System.out.println("Drawing a generic shape");

    }

    double calculateArea() {

        System.out.println("Calculating Generic Area...");//shan,72387,BSAI

        return 0;

    }

}
```

Class Circle

```
package com.mycompany.shapes.project;

public class Circle extends Shape {

    void draw() {

        System.out.println("Drawing a Circle");//shan,72387,BSAI

    }

}
```

Class Cylinder

```
public Cylinder(double r, double h) {

    this.radius = r;

    this.height = h;//shan,72387,BSAI

}

double calculateArea() {

    double area = (2 * Math.PI * radius * height) + (2 * Math.PI * radius * radius);

    return area;//shan,72387,BSAI

}}
```

Class Main

```
package com.mycompany.shapes.project;

import java.util.Scanner;

public class ShapesProject {

    public static void main(String[] args) {

        Scanner s = new Scanner(System.in);

        Circle c = new Circle();//shan,72387,BSAI

        c.draw();

        System.out.print("Enter radius for Cylinder: ");

        double r = s.nextDouble(); //shan,72387,BSAI

        System.out.print("Enter height for Cylinder: ");

        double h = s.nextDouble(); //shan,72387,BSAI

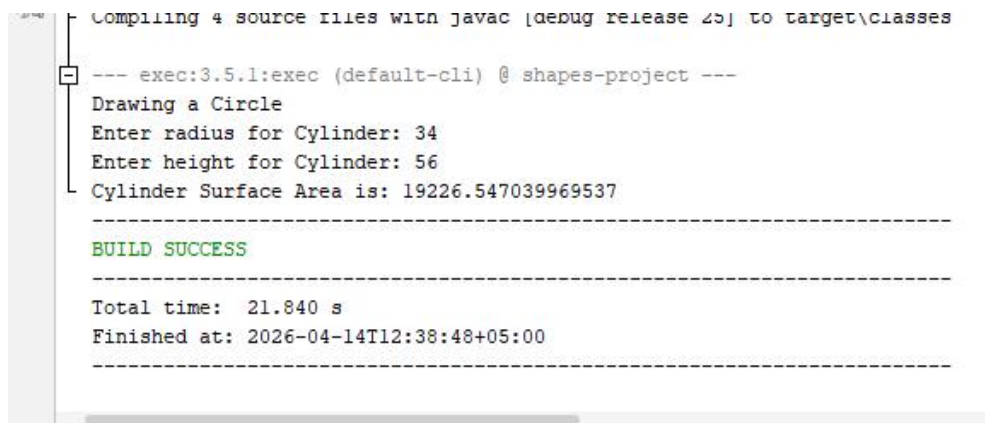
        Cylinder cyl = new Cylinder(r, h); //shan,72387,BSAI

        System.out.println("Cylinder Surface Area is: " + cyl.calculateArea());

    }}


```

➤ Output



```
Compiling 4 source files with javac [debug release 25] to target\classes
--- exec:3.5.1:exec (default-cli) @ shapes-project ---
Drawing a Circle
Enter radius for Cylinder: 34
Enter height for Cylinder: 56
Cylinder Surface Area is: 19226.547039969537
-----
BUILD SUCCESS
-----
Total time: 21.840 s
Finished at: 2026-04-14T12:38:48+05:00
-----
```

- ✧ **Q3:Write a Java program to create a base class BankAccount with methods deposit() and withdraw(). Create two subclasses SavingsAccount and CheckingAccount. Override the withdraw() method in each subclass to impose different withdrawal limits and fees.**

➤ **Input**

Class bankaccounts

```
package com.mycompany.bankaccount;

public class bankaccounts {

    double balance = 1000;//shan,72387,BSAI

    void deposit(double amount) {

        balance = balance + amount;//shan,72387,BSAI

        System.out.println("Balance: " + balance);

    }

    void withdraw(double amount) {

        System.out.println("Withdraw from BankAccount");//shan,72387,BSAI

    }

}
```

Class SavingAccount

```
package com.mycompany.bankaccount;

public class SavingsAccount {

    void withdraw(double amount) {

        if (amount > 500) {

            System.out.println("Limit exceeded");//shan,72387,BSAI

        } else {

            double balance = 0;

            balance = balance - amount;//shan,72387,BSAI

            System.out.println("Savings Balance: " + balance);

        }

    }

}
```

Class CheckingAccount

```

package com.mycompany.bankaccount;

public class CheckingAccount {

    void withdraw(double amount) {

        double balance = balance - amount - 10; //shan,72387,BSAI

        System.out.println("Checking Balance: " + balance);

    }

}

```

Class BankAccount(Main)

```

package com.mycompany.bankaccount;

public class BankAccount {

    public static void main(String[] args) {

        System.out.println("Testing Savings Account ");
        SavingsAccount s = new SavingsAccount();//shan,72387,BSAI
        s.deposit(1639);
        s.withdraw(728); //shan,72387,BSAI

        System.out.println("Testing Checking Account ");
        CheckingAccount c = new CheckingAccount();
        c.deposit(826);
        c.withdraw(452); //shan,72387,BSAI

    }

}

```

➤ Output

Balance: 1200

Savings Balance: 900

Balance: 1200

Checking Balance: 890